

7	(a)	(i)	$g = \frac{GM}{r^2}$ $g_M = \frac{G(0.11M_E)}{(0.53R_E)^2}$ $= \frac{0.11}{0.53^2} g_E$ $= 0.3916(9.81) = 3.84 \text{ m s}^{-2}$ [1] for correct subs to formula [1] for correct substitution by 9.81 m s⁻²	1
		(ii)	$\frac{GMm}{r^2} = mr\omega^2$ $GM = r^3 \left(\frac{2\pi}{T} \right)^2$ $\frac{r_E^3}{T_E^2} = \frac{r_M^3}{T_M^2}$ $\frac{T_M}{T_E} = \sqrt{\left(\frac{r_M}{r_E} \right)^3}$ $= \sqrt{1.52^3}$ $T_M = \sqrt{1.52^3} T_E = 1.87 \text{ yr}$	1
	(b)		$s = vt = (3.00 \times 10^8)(11 \times 60) = 1.98 \times 10^{11} \text{ m} = 1.98 \times 10^8 \text{ km}$	1
	(c)		$\text{Range} = 20 - (-140) = 160 \text{ }^\circ\text{C} = 160 \text{ K}$	1
	(d)		The air molecules gaining downward momentum implies the parachute is exerting a downward force on them, by Newton's 2nd law. The air molecules exert an equal but opposite upward force on the parachute, by Newton's 3rd law.	1
	(e)	(i)	$8F = Mg$ $8F = (1070 + 1025)(3.8)$ $F = 1000 \text{ N}$	1 1
		(ii)	$F = \frac{dp}{dt}$ $F = v \frac{dm}{dt}$ $1000 = v(1.7)$ $v = 588 \text{ m s}^{-1}$	1 1
		(iii)	$F_{\text{net}} = ma$ $(1025)(3.8) = (1070)a$ $a = 3.64 \text{ m s}^{-2}$	1 1

(f) (i)
$$\text{Efficiency} = \frac{110}{4100 \times 0.57} = 4.71\%$$

1

COMMENT: RTG (radioisotope thermoelectric generators), though very reliable and long-lasting, are very inefficient; **efficiencies** above 10% have never been achieved and most RTGs have efficiencies of 3–7%.
COMMENT: The rest of the heat generated is used to keep the rovers tools and systems at their correct operating temperature.

<https://mars.nasa.gov/mars2020/spacecraft/rover/electrical-power/>

(ii) **NOT GRADED!**

(iii) **NOT GRADED!**

(g) Advantage: Gravity on Mars is lower, so a smaller lift is required.

1

Disadvantage: Mars' air is thinner, so a larger rotor is required to generate the same lift.

1

[Total: 18]

~ End of paper ~